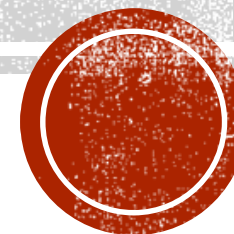


NATURAL LANGUAGE PROCESSING



WHAT IS NLP? AND WHY

- Natural Language Processing (NLP) is a branch of AI that enables computers to understand, interpret, and generate human language.
- **Why is NLP important?**
 - Automates text processing (e.g., emails, documents, social media)
 - Powers conversational AI (chatbots, virtual assistants)
 - Improves search engines (Google, Bing, etc.)
 - Enables language translation (Google Translate)



TEXT PRE-PROCESSING

- **Key Pre-processing Steps:**

- **Lowercasing:** Convert text to lowercase ("Hello" → "hello")
- **Removing Punctuation:** "Hello, world!" → "Hello world"
- **Stopword Removal:** Remove common words ("the", "is", "in")
- **Lemmatization:** Convert words to root form ("running" → "run")
- **Stemming:** Reduce words to base form ("playing" → "play")
- **Tokenization:** Split text into words or sentences

Python (NLTK):



WORD REPRESENTATION TECHNIQUES

- **One-Hot Encoding:** One-Hot Encoding is one of the **simplest ways to represent words numerically**. Each word in the vocabulary is represented by a **binary vector** where only **one position is 1 and the rest are 0**.

Word	Index	Word	Vector
cat	1	cat	[1,0,0]
dog	2	dog	[0,1,0]
apple	3	apple	[0,0,1]



WORD REPRESENTATION TECHNIQUES

- **Bag-of-Words (BoW):** Bag-of-Words represents text by **counting the frequency of each word** in a document.

It ignores **word order** but captures **how often words appear**.

"I love NLP and I love AI"

Word	Count	Vector: [2,2,1,1]
I	2	
love	2	
NLP	1	
AI	1	



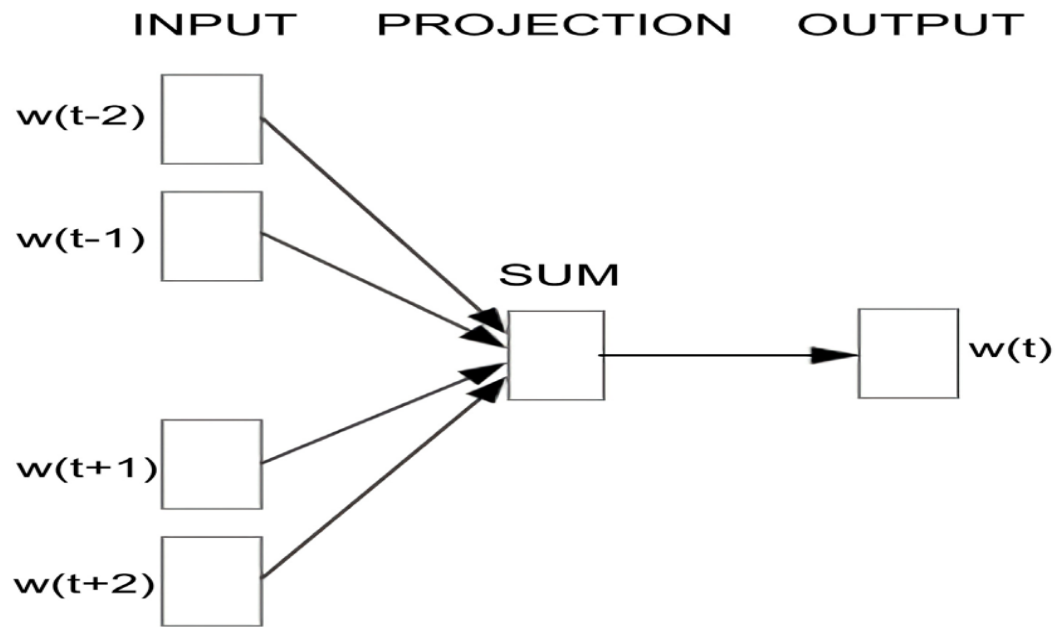
WORD REPRESENTATION TECHNIQUES

- **TF-IDF (Term Frequency – Inverse Document Frequency)**
- TF-IDF improves Bag-of-Words by assigning **importance weights** to words. Common words like "**the**", "**is**", "**and**" receive lower importance.
- **Term Frequency (TF)**
 - Measures how often a word appears in a document.
 - $TF = \text{Number of times a term appears} / \text{Total number of words}$
- **Inverse Document Frequency (IDF)** Reduces the importance of common words across documents. And IDF increases when a word is **rare across documents**.
- **Example Corpus:**
 - Document 1: "I love NLP"
 - Document 2: "I love machine learning"
- Word "**love**" appears in both documents → lower weight.
- Word "**NLP**" appears once → higher weight.

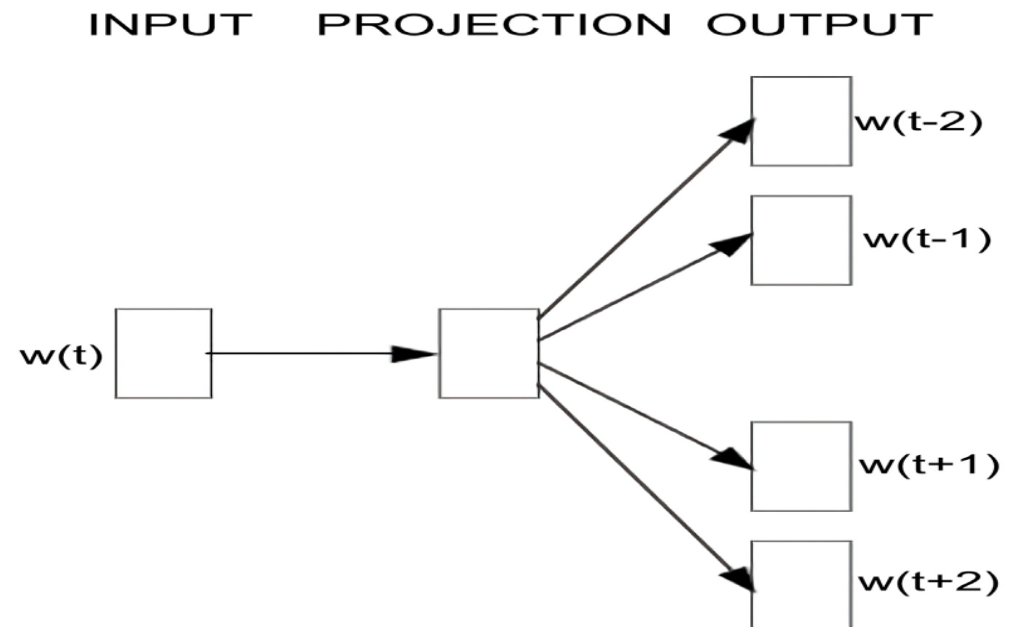


WORD REPRESENTATION TECHNIQUES

- **Word2Vec** to learn **word embeddings**



CBOW



Skip-gram



WORD REPRESENTATION TECHNIQUES

- **CBOW:** predict the **word from its context**
 - If the target word = sits,
 - Context: The cat ____ on the mat
 - The model uses the surrounding words:
 - [The, cat, on, the, mat] to predict:
 - Target: sits
- **Skip-Gram:** predict the **context from the word**

Context Words → Target Word

cat on
 \
 /
sits

Target Word → Context Words

sits
 /
 \
cat on



WORD REPRESENTATION TECHNIQUES

- **5. GloVe (Global Vectors)** GloVe combines **global word co-occurrence statistics** with embedding learning.
- It uses **matrix factorization** on a **word co-occurrence matrix**.
- Words that **appear together frequently** should have similar vectors.

- **Example:**

- ice → cold
steam → hot

- **How it Works**

- Build a **co-occurrence matrix**.

- Train vectors so their **dot product reflects word co-occurrence probabilities**.

<https://www.youtube.com/watch?v=9S0-OC4LFNo>

Probability and Ratio	$k = solid$	$k = gas$	$k = water$	$k = fashion$
$P(k ice)$	1.9×10^{-4}	6.6×10^{-5}	3.0×10^{-3}	1.7×10^{-5}
$P(k steam)$	2.2×10^{-5}	7.8×10^{-4}	2.2×10^{-3}	1.8×10^{-5}
$P(k ice)/P(k steam)$	8.9	8.5×10^{-2}	1.36	0.96

Very small or large:

solid is related to ice but not steam, or
gas is related to steam but not ice

close to 1:

water is highly related to ice and steam, or
fashion is not related to ice or steam.



WORD REPRESENTATION TECHNIQUES

"I love this movie"

- Step 1: Convert to tokens

["I", "love", "this", "movie"]

- Step 2: Convert to GloVe vectors

I → [0.13, 0.24, ...]

love → [0.85, 0.10, ...]

this → [0.40, 0.20, ...]

movie → [0.55, 0.90, ...]

Text



Tokenization



Word Index Mapping



Embedding Layer (GloVe vectors)



Neural Network (CNN / RNN /
Transformer)

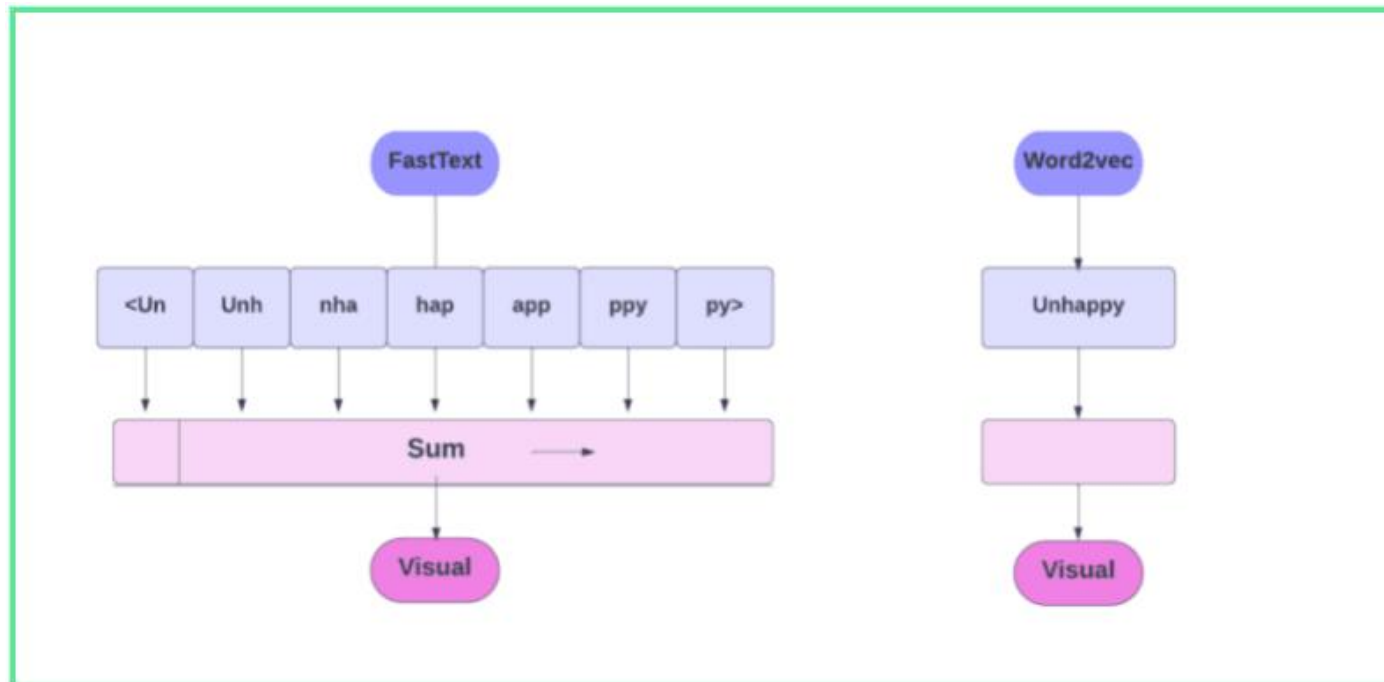


Prediction



WORD REPRESENTATION TECHNIQUES

- FastText not only considers whole words but also incorporates sub-word information like n-grams.
- 3-gram: "<bo", "boo", "ook", "oks", "ksh", "she", "hel", "elf", "lf>"

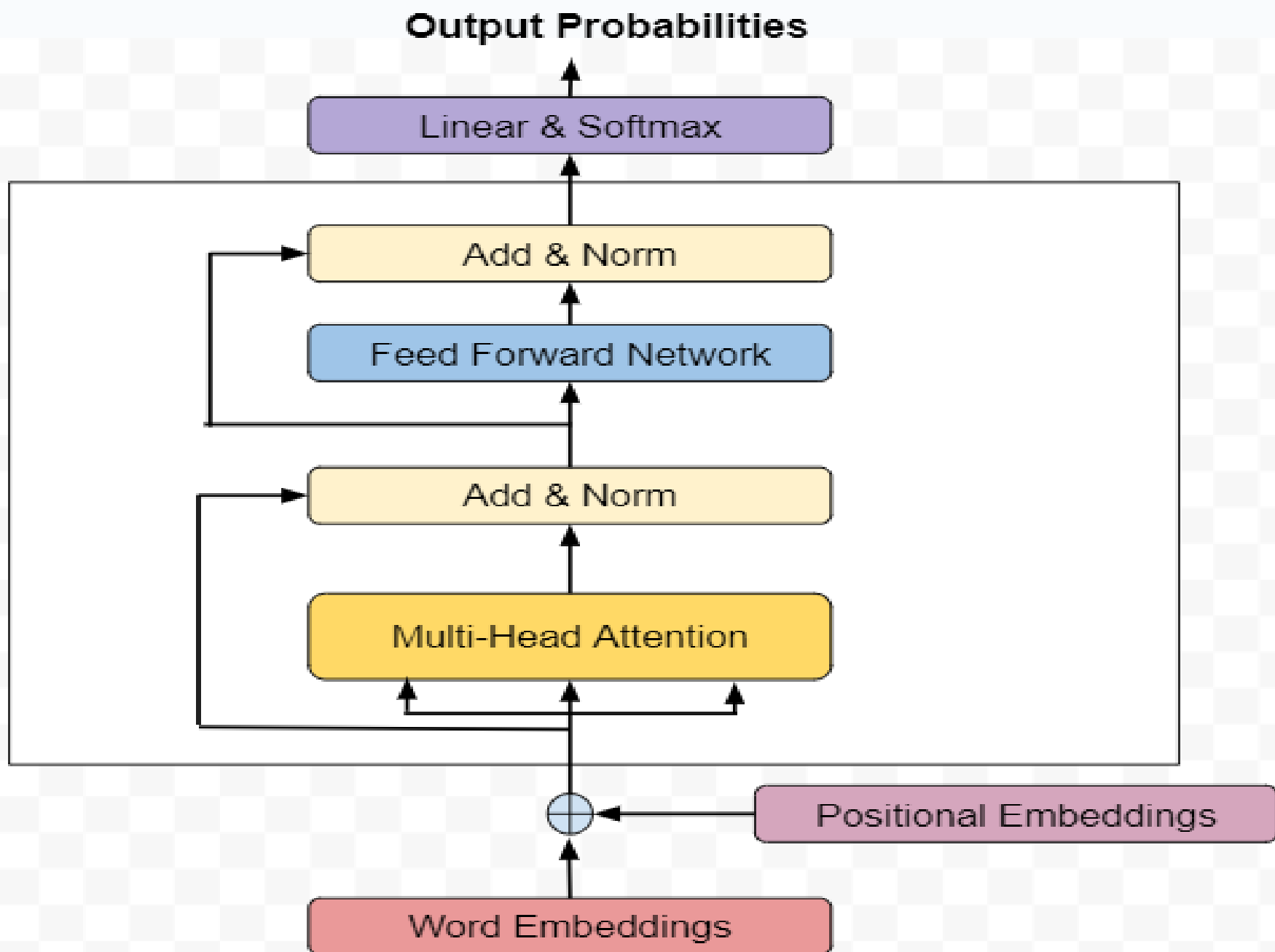


WORD REPRESENTATION TECHNIQUES

- **Bidirectional encoder representations from transformers (BERT)** produces **context-dependent word embeddings**.
- A word's meaning changes depending on the **sentence context**.
- Sentence 1: "I deposited money in the **bank**."
- Sentence 2: "The fisherman sat on the **bank**."
- **Attention mechanisms**
- BERT uses attention mechanisms to calculate how much each word should consider every other word in a sentence. For example, when BERT processes the word "bank," it assigns attention scores to all other words in that sentence. If "river" and "water" appear, they receive high scores, indicating that "bank" probably refers to a riverbank. If "money" and "deposit" score higher, BERT understands "bank" means a financial institution.
- https://www.youtube.com/watch?v=t45S_MwAcOw



12 ✕



Technique	Description	Example
One-Hot Encoding	Each word is represented as a sparse vector with a single 1 and the rest 0s. Simple but high-dimensional and cannot capture similarity between words.	"cat" → [0,0,1,0,...]
Bag-of-Words (BoW)	Represents a document by counting word occurrences, ignoring order. Good for simple text classification.	"I love NLP" → {"I":1, "love":1, "NLP":1}
TF-IDF (Term Frequency-Inverse Document Frequency)	Similar to BoW but weighs words by importance across documents. Reduces the impact of common words.	"I love NLP" in corpus → {"I":0.2, "love":0.5, "NLP":0.7}



Technique	Description	Example
Word Embeddings	Dense vector representations capturing semantic meaning. Words with similar meanings are close in vector space.	Word2Vec : "king" – "man" + "woman" ≈ "queen" GloVe : pretrained vectors from global co-occurrence
Contextualized Embeddings	Represent words based on context (dynamic vectors). Solves polysemy (words with multiple meanings).	BERT : "bank" in "river bank" vs "bank" in "money bank" → different vectors GPT embeddings for downstream tasks
Subword / Character-level Embeddings	Breaks words into subwords or characters to handle rare words or morphology.	FastText : "running" → embeddings of ["run", "ning"]



Metric Category	Purpose	Examples
Classification Metrics	Measure prediction quality in tasks like sentiment analysis, topic classification.	Accuracy, Precision, Recall, F1-score, ROC-AUC
Generation Metrics	Evaluate the quality of generated text, e.g., in summarization, translation.	BLEU, ROUGE, METEOR, CIDEr
Probabilistic Metrics	Assess the confidence and calibration of model predictions.	Perplexity, Log-likelihood, Brier score
Retrieval and Ranking Metrics	Evaluate search or recommendation performance.	Mean Average Precision (MAP), Normalized Discounted Cumulative Gain (NDCG), Precision@k, Recall@k
Agreement Metrics	Measure annotation quality or inter-annotator consistency in labeled datasets.	Cohen's Kappa, Fleiss' Kappa, Krippendorff's Alpha



Word Representation	Description	Typical NLP Tasks	Suitable Evaluation Metrics	Example
One-Hot Encoding	Sparse binary vector, 1 for word, 0 elsewhere	Text classification, simple NLP tasks	Classification Metrics: Accuracy, F1-score	"cat" → [0,0,1,0,...]
Bag-of-Words (BoW)	Counts of words in a document	Classification, spam detection, sentiment analysis	Classification Metrics: Accuracy, Precision, Recall, F1	"I love NLP" → {"I":1, "love":1, "NLP":1}
TF-IDF	Weighted word counts considering document frequency	Text classification, document retrieval	Classification Metrics, Retrieval & Ranking Metrics: MAP, NDCG	"I love NLP" → {"I":0.2, "love":0.5, "NLP":0.7}



Word Representation	Description	Typical NLP Tasks	Suitable Evaluation Metrics	Example
Word2Vec / GloVe (Static Embeddings)	Dense vectors capturing semantic similarity	Text classification, similarity, clustering	Classification Metrics, Probabilistic Metrics (if used in LM), Word Similarity Tasks: cosine similarity	"king" – "man" + "woman" $\approx$ "queen"
Contextual Embeddings (BERT, GPT, RoBERTa)	Dynamic vectors based on context	Question answering, sentiment analysis, NER, summarization	Classification Metrics, Generation Metrics (BLEU, ROUGE), Probabilistic Metrics (Perplexity)	"bank" in "river bank" vs "money bank" $\rightarrow$ different vectors
Subword / Character-level Embeddings (FastText)	Handles morphology & rare words	Text classification, language modeling	Classification Metrics, Probabilistic Metrics, Retrieval & Ranking Metrics	"running" $\rightarrow$ ["run", "ning"]



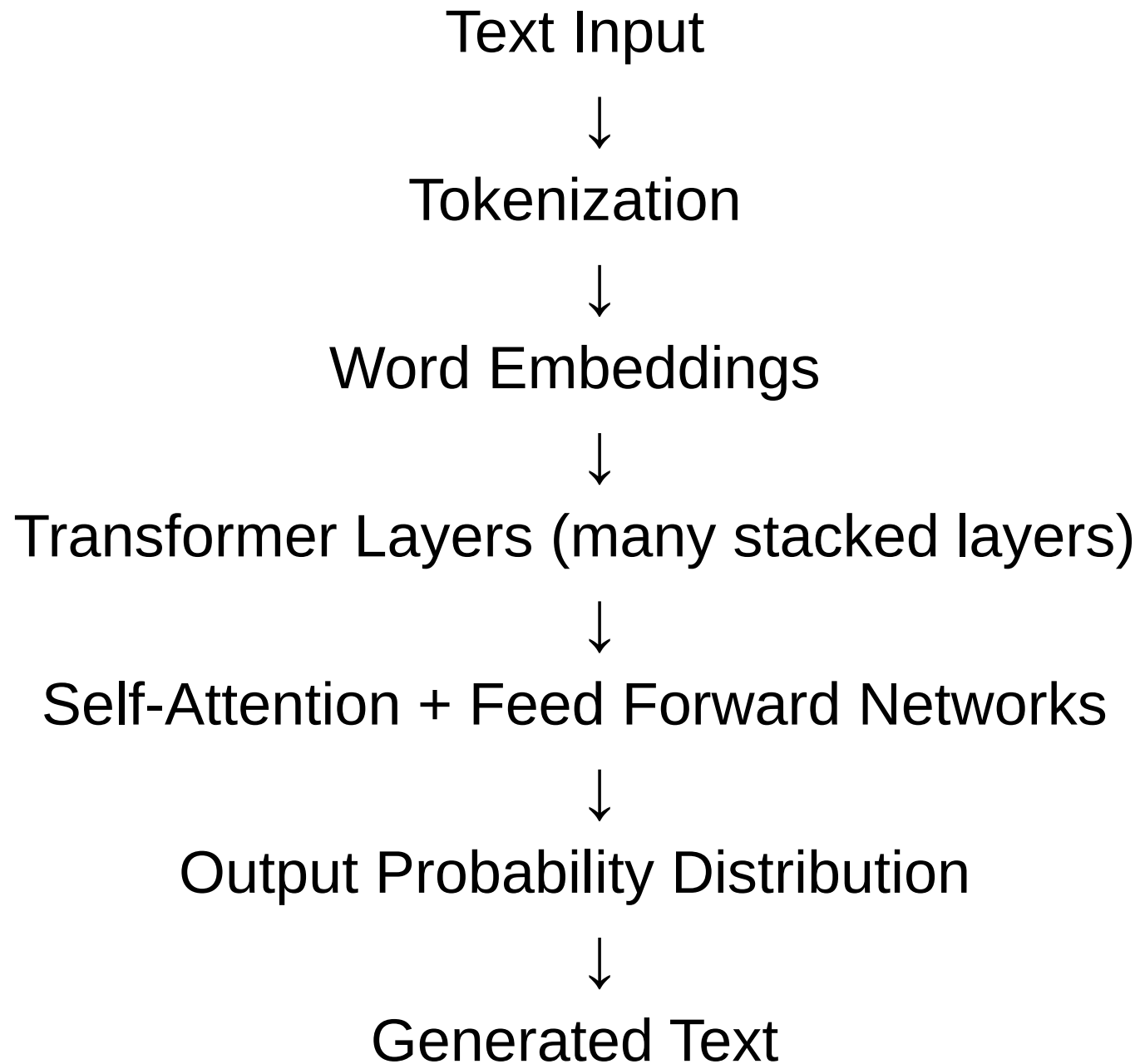
Linguistic Level	What It Means	Example	NLP Steps Related
Syntax	The grammatical structure of sentences	“The cat sat on the mat.” follows subject–verb–object structure	Text preprocessing, tokenization, POS tagging, parsing
Semantics	The meaning of words and sentences	“bank” can mean a financial institution or river edge	Word representations (Word2Vec, BERT), embeddings capturing meaning
Pragmatics	Meaning depending on context or intention	“Can you open the window?” → a request, not a question about ability	Contextual models, dialogue systems, chatbots, large language models



MODEL TRAINING

- Logistic Regression
- Naïve Bayes
- SVM
- CNN / RNN
- Transformers (BERT, GPT)





Simplified Architecture of an LLM

